

KodetoCareer — Implementation Plan

Version 2.0 | Updated — Free Infrastructure Stack

Only the changed sections are included here. All sprint content not listed below remains identical to Implementation Plan v1.

CHANGED SECTION: Pre-Sprint Preparation (Week 0)

Infrastructure & Tooling Setup (DevOps + Backend):

- **Oracle Cloud account** — sign up at cloud.oracle.com (no credit card required for Always Free tier)
- **Provision Oracle Cloud VM:**
 - Shape: VM.Standard.A1.Flex
 - OCPUs: 4, RAM: 24GB, Storage: 200GB boot volume
 - OS: Ubuntu 22.04 LTS
 - All permanently free, never expires
- Install on VM: Node.js 20, Nginx, PM2, Git, Certbot (not needed if using Cloudflare SSL)
- Configure Nginx as reverse proxy to port 3000
- Open VM security group ports: 22 (SSH), 80 (HTTP), 443 (HTTPS)
- **Neon.tech account** — sign up at neon.tech (free, no credit card required)
 - Create project: "kodetocareer-production"
 - Copy connection string → set as DATABASE_URL
 - Enable connection pooling in Neon dashboard
- **Upstash account** — sign up at upstash.com (free, no credit card required)
 - Create Redis database: "kodetocareer-cache"
 - Copy REST URL and Token → set as UPSTASH_REDIS_* env vars
 - Copy Redis URL → set as UPSTASH_REDIS_URL (for Bull queue)
- **Cloudflare account** — sign up at cloudflare.com (free)
 - Add your domain → update nameservers at your registrar
 - Enable SSL/TLS: Full (strict) mode
 - Create R2 bucket: "kodetocareer-production"
 - Create R2 API token with read/write permissions
 - Add custom domain to R2 bucket: cdn.kodetocareer.com
 - ■ This replaces: AWS S3 + CloudFront + AWS Certificate Manager
- **Resend account** — sign up at resend.com (free, no credit card required)
 - Verify your sending domain (kodetocareer.com)
 - Create API key → set as RESEND_API_KEY
 - Free tier: 3,000 emails/month, 100/day

- ■ This replaces: AWS SES
- **Firestore project** — console.firebase.google.com (free)
 - Enable Cloud Messaging (FCM) for push notifications
 - Enable Analytics and Crashlytics for mobile app
- ■ No change from original
- **Sentry account** — sentry.io (free tier: 5,000 errors/month)
 - Create two projects: kodetocareer-api, kodetocareer-app
- **UptimeRobot** — uptimerobot.com (free: 50 monitors)
 - Add monitor for: api.kodetocareer.com/health
 - Set alert email for downtime notification
- **GitHub repository** — github.com (free for private repos)
 - Create repo, set up branch strategy (main, develop, feature/*)
 - Add GitHub Secrets: ORACLE_HOST, ORACLE_SSH_KEY
 - Set up GitHub Actions CI/CD workflow (free 2,000 min/month)
- **Domain registration**
 - Register domain at Namecheap or GoDaddy (~■800/year — only non-free item)
 - Point nameservers to Cloudflare
- Database migration: `npx prisma migrate deploy` against Neon connection string
- Seed data: 2 pilot colleges, 5 batches, 50 test students

REMOVED from original Week 0 (all were paid):

- ~~AWS account setup~~ → Oracle Cloud (free)
- ~~S3 bucket creation~~ → Cloudflare R2 (free)
- ~~RDS PostgreSQL~~ → Neon.tech (free)
- ~~ElastiCache Redis~~ → Upstash (free)
- ~~CloudFront distribution~~ → Cloudflare CDN (free)
- ~~AWS Certificate Manager~~ → Cloudflare SSL (free)
- ~~SES configuration~~ → Resend (free)
- ~~MSC01 configuration~~ → Email OTP only (free)

CHANGED SECTION: Sprint 4 — Video Delivery (Week 4)

REMOVED (was paid AWS MediaConvert pipeline):

- ~~MediaConvert job creation~~
- ~~Lambda function for processing~~
- ~~HLS transcoding (480p/720p/1080p)~~
- ~~Adaptive bitrate streaming~~

REPLACED WITH (free):

Backend:

- Direct MP4 upload via R2 presigned URL (same SDK, S3-compatible)
- On upload confirm: update lesson.videoStatus = 'READY', store R2 CDN URL
- No transcoding needed — video served directly from Cloudflare CDN
- YouTube/Vimeo embed URL support (paste URL, store as lesson.external_url)
- Thumbnail: extract first frame client-side using canvas API (or let admin upload manually)

Flutter:

- Video player integration (video_player + chewie)
- For direct uploads: play MP4 from Cloudflare CDN URL
- For YouTube/Vimeo: use webview_flutter to embed (or youtube_player_flutter package)
- Custom controls: speed, fullscreen, PiP
- Video progress persistence
- "Mark complete" button (after 80% watched)

React (Admin):

- Video source type selector: [Upload MP4] vs [YouTube/Vimeo URL]
- Upload: presigned R2 URL flow with progress bar
- Status: once upload confirmed, status immediately becomes READY (no processing wait)
- URL embed: paste YouTube/Vimeo URL → preview shown inline

Sprint 4 Deliverable: Full video lesson delivery working — both direct upload (MP4 on R2) and YouTube/Vimeo embed. No MediaConvert dependency.

CHANGED SECTION: Sprint 9 — SMS Change

Sprint 9 — Communication (Original line):

~~— [] SMS integration (MSG91 for OTP and alerts)~~

Replace with:

- Email OTP system (already built in Sprint 1 — no change needed)
- All critical alerts via email (Resend) — attendance below threshold, quiz result, assignment graded
- Note in system settings: SMS available as future paid add-on (Twilio/MSG91) when business generates revenue

No MSG91 or Twilio account needed.

CHANGED SECTION: Risk Register Updates

Risks REMOVED (no longer applicable):

- ~~MediaConvert costs exceed budget~~ — removed from stack
- ~~AWS RDS pricing~~ — replaced with Neon.tech (free)

- ~~Push notifications not delivered~~ — FCM still used, no change

Risks UPDATED:

Risk | Likelihood | Impact | Mitigation

Neon.tech 0.5GB storage limit | Low (MVP) | Medium | 0.5GB holds ~50,000 student records easily. Upgrade to Neon paid (\$19/month) only when needed

Upstash 10K commands/day limit | Medium (at scale) | Low | At 1,000 daily users: ~10 commands/user = 10,000. Monitor and upgrade when needed (\$0.20/100K commands)

Cloudflare R2 10GB storage limit | Low (MVP) | Medium | 10GB = ~200 videos at 50MB average. Use YouTube embed for large videos to stay within limit

Oracle Cloud VM single instance | Medium | High | PM2 cluster mode uses all 4 OCPUs. Take daily DB snapshot. Recovery < 1 hour

Resend 100 emails/day limit | Medium (at scale) | Medium | 100/day = 3,000/month. For 500 students this is fine. Upgrade to Resend paid (\$20/month) at scale

Video quality without transcoding | Low | Low | For MVP, one quality level is acceptable. Add transcoding (Cloudflare Stream) in v2 when revenue exists

Oracle Cloud account free tier limit | Low | High | Oracle has been stable with free tier. No reports of forced upgrades. Keep account active with monthly login

CHANGED SECTION: Budget Estimate

Infrastructure Cost (Updated — Free Stack)

Service | Cost | Notes

Oracle Cloud VM (4 OCPU, 24GB RAM) | ****0**** | Always Free, permanent

Neon.tech PostgreSQL | ****0**** | Free tier, permanent

Upstash Redis | ****0**** | Free tier, 10K cmd/day

Cloudflare R2 Storage (10GB) | ****0**** | Free tier, permanent

Cloudflare CDN + SSL + DNS | ****0**** | Free, permanent

Resend Email (3,000/month) | ****0**** | Free tier, permanent

Firebase FCM + Analytics | ****0**** | Free, permanent

Sentry (5,000 errors/month) | ****0**** | Free tier, permanent

UptimeRobot (50 monitors) | ****0**** | Free tier, permanent

GitHub Actions CI/CD | ****0**** | Free for private repos

Domain (kodetocareer.com) | ****~800/year**** | Only unavoidable cost

MSG91 / Twilio SMS | ****0**** | Removed from MVP

****Total Monthly Infrastructure**** | ****0/month**** | (**67/month** amortised domain)

Compared to original plan: saves ■31,300/month = ■3,75,600/year

When to Upgrade (Paid Services)

Only upgrade when you have revenue. These are the thresholds:

When This Happens | Then Add

>3,000 emails/month | Resend paid: \$20/month

>10,000 Redis commands/day | Upstash pay-per-use: ~\$2/month

>10GB video storage on R2 | Use YouTube unlisted (free) OR Cloudflare Stream (\$5/1,000 min)

>0.5GB database | Neon paid: \$19/month (scales to 100GB)

SMS OTP needed | Twilio or MSG91: ~■1,500/month

End of Updated Implementation Plan Sections

All other sprint content (1, 2, 3, 5, 6, 7, 8, 10) is unchanged from v1.